is a cocartesian diagram, with f open. In particular f is a monomorphism, and it follows that f' is a monomorphism and that the square is cartesian as well. The morphism $F' \coprod G \to G'$ is onto. The pull-back of f' by $f' : F' \to G'$ is an isomorphism (f' being a monomorphism) hence open. The pull-back of f' by $G \to G'$ is just f, open by assumption. It follows that f' is open. $\square$

We now fix a class C of open embeddings $U \to V$ in (QP/G). We require the following stabilities

Conditions 34. 1. If $U \to U' \to V$ are open embeddings and if $U \to V$ is in C, so is $U' \to V$.

2. If $U \to V$ is an open embedding in C, and if $f : V' \to V$ is etale, with $f^{-1}(Y) \subset Y$ for Y the complement of U in V, then $f^{-1}(U) \to V'$ is in C.

The classes C we will have to consider are the following:

1. The open embeddings $U \to V$ with V smooth.
2. The open embeddings $U \to V$ with V smooth such that the action of G is free on $V - U$. Equivalently: V is the union of U and the open subset on which the action of G is free.
3. When working in (Sm/S): all open embeddings.

Definition 7. A morphism $f : F \to G$ is C-solid if it is a composite $F = F_0 \to F_1 \to \ldots \to F_n = G$ where each $F_i \to F_{i+1}$ is deduced by push-out from some $h_U \to h_X$, $U \subset X$ in C.

A sheaf F is solid if $\emptyset \to F$ is C-solid.

In the pointed context, a pointed sheaf is (pointed) C-solid if the morphism $pt \to F$ is C-solid.

Example 1. For U open in X, let $h_{X/U}$ be the sheaf h_X, with the subsheaf h_U contracted to a point p. If $U \to X$ is in C, then $p : pt \to h_{X/U}$ is solid: it is the push-out of $h_U \to h_X$ by $h_U \to pt$. Thom spaces are of this form: starting from a vector bundle V on Y, one contracts, in the total space of this vector bundle, the complement of the zero section to a point.

The class of solid morphisms is the smallest class closed by compositions and push-outs which contains all $h_U \to h_X$ for $U \subset X$ in C. By Proposition 33 solid morphisms are open.

For $F \to G$ a monomorphism of sheaves, define G/F to be the pointed sheaf obtained by contracting F to a point: one has a cocartesian square

$$\begin{array}{ccc} F & \longrightarrow & G \\ \downarrow & & \downarrow \\ pt & \longrightarrow & F/G \end{array}$$

By transitivity of push-out, any cocartesian diagram

$$\begin{array}{ccc} F & \longrightarrow & G \\ \big\downarrow & & \big\downarrow \\ F' & \longrightarrow & G' \end{array}$$

induces an isomorphism $G/F \to G'/F'$.

Proposition 35. *A morphism of sheaves $f : F \to G$ is C-solid if and only if it is a composite $F = F_0 \to F_1 \to \ldots \to F_n = G$ of monomorphisms where each F_i/F_{i+1} is isomorphic to some $h_{V/U} = h_V/h_U$ for $U \subset V$ in C.*

Proof. If a morphism $F \to G$ is deduced by push-out from $U \to V$, G/F is isomorphic to $h_{V/U}$. From this, "only if" results. Conversely, if we have

$$\begin{array}{ccc} F & \longrightarrow & G \\ \big\downarrow & & \big\downarrow \\ * & \longrightarrow & h_{V/U} \end{array}, \quad \begin{array}{ccc} h_U & \longrightarrow & h_V \\ \big\downarrow & & \big\downarrow \\ * & \longrightarrow & h_{V/U} \end{array} \tag{35.1}$$

cocartesian, and if $h_V \to h_{V/U}$ lifts to G, then $F \to G$ is deduced by push-out from $h_U \to h_V$. Indeed, the diagrams (35.1) being cartesian as well as cocartesian, we have a cartesian

$$\begin{array}{ccc} h_U & \longrightarrow & h_V \\ \big\downarrow & & \big\downarrow \\ F & \longrightarrow & G \end{array}$$

If G_1 is deduced from $h_U \to h_V$ by push-out:

$$\begin{array}{ccccc} h_U & \longrightarrow & h_V & \xrightarrow{=} & h_V \\ \big\downarrow & & \big\downarrow & & \big\downarrow \\ F & \longrightarrow & G_1 & \longrightarrow & G \end{array}$$

then $G_1/F \cong G/F$ and it follows that $G_1 = G$.

Let us suppose only that we have $v : \tilde{V} \to V$ etale, inducing an isomorphism from $\tilde{V} - v^{-1}(U)$ to $V - U$ and a lifting of $h_{\tilde{V}} \to h_{V/U}$ to G. If $\tilde{U} := v^{-1}(U)$, $h_{\tilde{V}/\tilde{U}} \to h_{V/U}$ is an isomorphism. This is most easily checked by applying the fiber functors defined by a G-local henselian Y: a morphism $Y \to V$, if it does not map to U, lifts uniquely to a morphism to $\tilde{V}$. The assumptions made hence imply that $F \to G$ is a push-out of $h_{\tilde{U}} \to h_{\tilde{V}}$. Note that by the second stability property of C, $\tilde{U} \to \tilde{V}$ is in C.

We will reduce the proof of "if" to that case. We have to show that if a monomorphism $f : F \to G$ is such that $G/F \cong h_{V/U}$ with $U \to V$ in C, then f is C solid. The cocartesian square

$$\begin{array}{ccc} F & \longrightarrow & G \\ \downarrow & & \downarrow \\ pt & \longrightarrow & F/G \end{array} \tag{35.2}$$

induces an epimorphism $pt \coprod G \to h_{V/U}$. The natural section of $h_{V/U}$ on V can hence locally be lifted to pt or to G: for some filtration $\emptyset = Z_n \subset \cdots \subset Z_1 \subset Z_0 = V$ of V by closed equivariant subschemes, we have etale maps $\phi_i : Y_i \to V$ with a (equivariant) section over $Z_i - Z_{i+1}$, and a lifting of $h_{Y_i} \to h_{V/U}$ to pt or to G. Note $V_i := V - Z_{i+1}$. We may:

1. Start with $V_0 = U$, taking $Y_0 = U$: here the lifting is to pt
2. Assume $V_i \neq V_{i+1}$; the succeeding liftings then cannot be to pt: they must be to G
3. Shrink Y_i, first so that it maps to V_i, next so that it induces an isomorphism from $Y_i - \phi_i(V_{i-1})$ to $Z_i - Z_{i+1}$

As $F \to G$ is a monomorphism the cocartesian (35.2) is cartesian as well. The composition

$$pt = h_{V_0/U} \to h_{V_1/U} \to \ldots \to h_{V/U}$$

gives by pull-back a factorization of $F \to G$ as

$$F \to F_1 \to \ldots \to G$$

with each

$$\begin{array}{ccc} F_i & \longrightarrow & F_{i+1} \\ \downarrow & & \downarrow \\ h_{V_i/U} & \longrightarrow & h_{V_{i+1}}/U \end{array}$$

cartesian and cocartesian, hence $F_{i+1}/F_i \cong h_{V_{i+1}}/h_{V_i}$. Further, the morphism $\phi_{i+1} : h_{Y_{i+1}} \to h_{V_{i+1}} \to h_{V/U}$ lifts to G, hence $h_{Y_{i+1}} \to h_{V_{i+1}/U}$ lifts to F_{i+1}. It follows that $F_i \to F_{i+1}$ is a push-out of $\phi_{i+1}^{-1}(V_i) \to Y_{i+1}$, which is in C, and solidity follows. □

Remark 1. Another formulation of Proposition 35 is: a morphism $F \to G$ is C-solid if and only if the pointed sheaf G/F is an iterated extension of $h_{V/U}$'s with $U \to V$ in C, in the sense that there are morphisms

$$pt = H_0 \to \ldots \to H_n = G/F$$

with each H_{i+1}/H_i of the form $h_{V/U}$.

Proposition 36. *If $f : F \to G$ is open and G is C-solid, then f is C-solid.*

Proof. In the proof we say "solid" instead of "C-solid". Let (*) be the property of a sheaf G that any open $f : F \to G$ is solid. If G is solid, G sits at the end of a chain $\emptyset = G_0 \to G_1 \to \ldots \to G_n = G$ with each $G_i \to G_{i+1}$ push out of some $h_U \to h_X$ for $U \to X$ in C. We prove by induction on i that G_i satisfies (*).

For $i = 1$, $G_1 = h_X$ is representable and $\emptyset \to X$ is in C. If $f : F \to G_1$ is open, it is of the form $h_U \to h_X$ for U open in X, hence solid by Condition 34(1). It remains to check that if in a cocartesian square

$$\begin{tikzcd} h_U \arrow[r] \arrow[d] & h_X \arrow[d] \\ G' \arrow[r] & G \end{tikzcd} \tag{36.1}$$

the sheaf G' satisfies (*), so does G. In (36.1), $h_U \to h_X$ is a monomorphism and the square (36.1) hence cartesian as well as cocartesian.

Fix $f : F \to G$ open, and take the pull-back of (36.1) by f. It is again a cartesian and cocartesian square and, f being open, it is of the form

$$\begin{tikzcd} h_V \arrow[r] \arrow[d] & h_Y \arrow[d] \\ F' \arrow[r] & F \end{tikzcd} \tag{36.2}$$

where Y is open in X and $V = U \cap Y$. The diagram

$$\begin{tikzcd} F' \arrow[r] \arrow[d] & F \arrow[r] \arrow[d] & h_{Y/V} \arrow[d] \\ G' \arrow[r] \arrow[d] & G \arrow[r] \arrow[d] & h_{X/U} \arrow[d] \\ G'/F' \arrow[r] & G/F \arrow[r] & h_{X/(U \cup V)} \end{tikzcd}$$

expresses G/F as an extension of $h_{X/(U \cup Y)}$ by G'/F' and one concludes by Remark 1 using (*) for G' and the fact that $U \cup Y \to X$ is in C. $\square$

Proposition 37. *The pull-back of a solid morphism f by an open morphism s is solid. In particular, if $g : F \to G$ is open and if G is solid, so is F.*

Proof. Since the pull-back of an open morphism is open, it suffices to check the proposition for f a push-out of $h_U \subset h_X$ for U open in X:

$$\begin{tikzcd} h_U \arrow[r] \arrow[d] & h_X \arrow[d] \\ G' \arrow[r, "f"] & G \end{tikzcd}$$

Pulling back by g, we obtain a cocartesian square

$$\begin{tikzcd} h_{U'} \arrow[r] \arrow[d] & h_{X'} \arrow[d] \\ F' \arrow[r] & F \end{tikzcd}$$

with U' open in U and X' open in X. This shows that $F' \to F$ is solid. $\square$

Suppose now that we are given two classes C and C' of open embeddings satisfying conditions 34. We define $C \times C'$ as the smallest class stable by Condition 34 containing the

$$(U \times V') \cup (U' \times V) \subset V \times V'$$

for $U \subset V$ in C and $U' \subset V'$ in C'.

Example 2. If C is a class of all open embeddings and C' is the class of those $U' \subset V'$ for which G acts freely outside U', then $C \times C' = C'$.

Proposition 38. *If the pointed sheaves F and F' are respectively C and C'-solid, the $F \wedge F'$ is $C \times C'$-solid.*

Proof. By assumption, F is an iterated extension in the sense of Remark 1 of pointed sheaves h_{V_i/U_i} with $U_i \to V_i$ in C. Similarly for F', with $U'_j \to V'_j$ in C'. The smash product $F \wedge F'$ is then an iterated extension of the

$$h_{V_i/U_i} \wedge h_{V'_j/U'_j} = h_{V_i \times V'_j/((U_i \times V'_j) \cup (V_i \times U'_j))},$$

taken for instance in the lexicographical order, hence it is $C \times C'$ solid. $\square$

Definition 8. A morphism is called ind-solid relative to C if it is a filtering colimit of C-solid morphisms.

Exercise 39. We take G to be the trivial group. A section on Y of a push-out

$$\begin{tikzcd} h_U \arrow[r] \arrow[d, "\psi"'] & h_X \arrow[d] \\ F \arrow[r] & G \end{tikzcd}$$

can be described as follows. For an open subset V of Y and a section ϕ of F on V consider on the small Nisnevich site Y_{Nis} of Y the presheaf $\Phi(V, \phi)$ which sends $a : W \to Y$ to the set of morphisms $f : W \to X$ such that $f^{-1}(U) = a^{-1}(V)$ and $\phi_{|a^{-1}(V)} = f^*(\psi)$. A section of G on Y is given by data:

1. An open subset V of Y.
2. A section ϕ of F on V.

3. A section of $i^*(a_{Nis}\Phi(V,\phi))$ on $Y - V$ where i is the closed embedding $Y - V \to Y$ and a_{Nis} denotes the associated Nisnevich sheaf.

Exercise 40. In the notations of Exercise 39, if F is a sheaf for the etale topology, so is G. For any Y, the (V,ϕ) as in (1),(2) above form a sheaf for the etale topology. It hence suffices to prove that for (V,ϕ) fixed, the datum (3) forms a sheaf for the etale topology. This is checked by using the following criterion to check if a Nisnevich sheaf is etale. For $y \in Y$, and for L a finite separable extension of k_y, let $\mathcal{O}^h_{L,y}$ be deduced by "extension of the residue field" from the henselization $\mathcal{O}^h_y$ of Y at y. The criterion is that $Spec(L) \mapsto F(Spec(\mathcal{O}^h_{L,y}))$ should be an etale sheaf on $Spec(k_y)_{et}$.

Exercise 41. It follows from Exercises 39 and 40 that if $f : F \to G$ is ind solid, and if F is etale, then G is etale. In particular, a solid sheaf, as well as a pointed solid sheaf, are etale sheaves.

Remark 2. The same formalism of open and solid morphisms holds in the site of all schemes of finite type over S with the etale topology.

4.2 A Criterion for Preservation of Local Equivalences

We work with pointed sheaves on QP/G. Our aim in this section is to prove the following result

Theorem 6. *Let Φ be a functor from pointed sheaves to pointed sets. Suppose that Φ commutes with all colimits, and that for any open embedding $U \to X$, $\Phi((h_U)_+) \to \Phi((h_X)_+)$ is a monomorphism. Then if $f : F_\bullet \to G_\bullet$ is a local equivalence and if F_n and G_n are (pointed) ind-solid, then $\Phi(f)$ is a weak equivalence.*

Suppose that

$$Q \quad : \quad \begin{array}{ccc} B & \longrightarrow & Y \\ \downarrow & & \downarrow \\ A & \longrightarrow & X \end{array}$$

is an upper distinguished square. Adding a base point, we obtain Q_+. The morphism $K_{Q_+} \to X_+$ is then a local equivalence. Let us check that $\Phi(K_{Q_+}) \to \Phi(X_+)$ is a weak equivalence. As Φ commutes with coproducts, this morphism is deduced from the commutative square

$$\begin{array}{ccc} \Phi((h_B)_+) & \longrightarrow & \Phi((h_Y)_+) \\ \downarrow & & \downarrow \\ \Phi((h_A)_+) & \longrightarrow & \Phi((h_X)_+) \end{array}$$

by applying the same construction (23.3). This square is cocartesian because Q is. The top horizontal line being a monomorphism, it is homotopy cocartesian, and the claim follows. As Φ commutes with colimits, this special case implies that more generally one has

Lemma 15. *If f is in the $\bar{\Delta}$-closure of the $(K_Q)_+ \to X_+$ as above, then $\Phi(f)$ is a weak equivalence.*

Proposition 42. *(Proof of Theorem 6)* For any pointed sheaf F, $R(F) \to F$ is a local equivalence. Indeed for any connected X in QP/G, $R(F)(X) \to F(X)$ is a weak equivalence by Construction 29. It follows that for $f : F \to G$ a local equivalence,

$$\begin{tikzcd} R(F) \arrow[r] \arrow[d] & R(G) \arrow[d] \\ F \arrow[r] & G \end{tikzcd}$$

is a commutative square of local equivalences. By Lemmas 15 and 12, $\Phi(R(f))$ is a weak equivalence. It remains to see that $\Phi(R(F)) \to \Phi(F)$ is a weak equivalence – and the same for G. For this it suffices to see that for a pointed ind-solid sheaf F, $\Phi(R(F)) \to \Phi(F)$ is a weak equivalence. As Φ and R commute with filtering colimits, the ind-solid reduces to solid, and by the inductive definition of solid, it suffices to prove the following lemma.

Lemma 16. *Let $U \to X$ be an open embedding. If in a cartesian square of pointed sheaves*

$$Q \quad : \quad \begin{tikzcd} (h_U)_+ \arrow[r] \arrow[d] & (h_X)_+ \arrow[d] \\ F \arrow[r] & G \end{tikzcd}$$

F is such that $\Phi R(F) \to \Phi(F)$ is a weak equivalence, the same holds for G.

Proof. Consider the cocartesian square

$$Q' \quad : \quad \begin{tikzcd} R((h_U)_+) \arrow[r] \arrow[d] & R((h_X)_+) \arrow[d] \\ R(F) \arrow[r] & R \end{tikzcd}$$

One can easily see that the top morphism is a monomorphism. It follows that Q' is point by point homotopy cocartesian, and $R \to R(G)$ is a local equivalence. The functor i^*i_* of Construction 29 transforms a monomorphism into a coprojection of the form $A \to A \vee ((\coprod h_{U_i})_+)$. It follows that each R_n is of the form $(\coprod h_{U_i})_+$ and, by Lemmas 15 and 12, $\Phi(R) \to \Phi(R(G))$ is a weak equivalence. It remains to show that $\Phi(R) \to \Phi(G)$ is a weak equivalence.

Let us apply Φ to the morphism of cocartesian squares $Q' \to Q$. By Construction 29 both $R((h_U)_+) \to (h_U)_+$ and $R((h_X)_+) \to (h_X)_+$ are homotopy equivalences, and remain so by applying Φ. We assumed $\Phi R(F) \to \Phi(F)$ to be a weak equivalence. As $\Phi(Q')$ and $\Phi(Q)$ are cocartesian with a top morphism which is a monomorphism (by the assumption on Φ, for Q), it follows that $\Phi(R) \to \Phi(G)$ is a weak equivalence. Hence so is $\Phi(R(G)) \to \Phi(G)$. □

5 Two Functors

5.1 The Functor $X \mapsto X/G$

One has a natural morphism of sites

$$\eta : (QP/G)_{Nis} \to (QP)_{Nis}$$

given by the functor

$$\eta^f : QP \to QP/G : X \mapsto (X \text{ with the trivial } G\text{-action})$$

Indeed, the functor η^f commutes with finite limits and transforms covering families into covering families.

In particular the functor η^f is continuous: if F is a sheaf on $(QP/G)_{Nis}$, the presheaf

$$X \mapsto F(X \text{ with the trivial } G\text{-action})$$

is a sheaf on $(QP)_{Nis}$. The functor η^f has a left adjoint $\lambda^f : X \mapsto X/G$. As η^f is continuous, the functor λ^f is cocontinuous, and the functor η^* from sheaves on $(QP)_{Nis}$ to sheaves on $(QP/G)_{Nis}$ is

$$F \mapsto (\text{sheaf associated to the presheaf } X \mapsto F(X/G))$$

Proposition 43. *The cocontinuous functor $\lambda^f : X \mapsto X/G$ is also continuous, that is, if F is a sheaf on $(QP)_{Nis}$, the presheaf $X \mapsto F(X/G)$ on $(QP/G)_{Nis}$ is a sheaf.*

Proof. By Lemma 2 it is sufficient to test the sheaf property of $X \mapsto F(X/G)$ for a covering of X deduced by pull-back from a Nisnevich covering $V_i \to X/G$ of X/G. Passage to quotient commutes with flat base change. Taking as base X/G, this gives that

$$X \times_{X/G} V_i \to V_i$$

identifies V_i with the quotient of $X \times_{X/G} V_i$ by G. Similarly, if $V_{ij} = V_i \times_{X/G} V_j$, the quotient by G of the pull-back to X of V_{ij} is V_{ij} again. This reduces the sheaf property of $X \mapsto F(X/G)$, for the covering of X by the $X \times_{X/G} V_i$, to the sheaf property of F for the covering (V_i) of X/G. □

The functor $\lambda^f : X \mapsto X/G$ gives rise to a pair of adjoint functors (λ_*, λ^*) between the categories of presheaves on (QP/G) and (QP), with $\lambda_*(F)$ being $X \mapsto F(X/G)$. As λ^f is continuous , it induces a similar pair of adjoint functors between the categories of sheaves. This pair is

$$(\eta_\# := (\text{associated sheaf}) \circ \lambda^*, \eta^* = \lambda_*),$$

so that one has a sequence of adjunctions $(\eta_\#, \eta^*, \eta_*)$. If F on (QP/G) is representable: $F = h_X$, then $\eta_\#(F) = h_{X/G}$. In particular, $\eta_\#$ transforms the final sheaf h_S on $(QP/G)_{Nis}$, also called "point", into the final sheaf on $(QP)_{Nis}$, and $(\eta_\#, \eta^*)$ is a pair of adjoint functors in the category of pointed sheaves as well. It is clear that $\eta_\#$ takes solid sheaves to solid sheaves. We also have the following.

Proposition 44. *Let F be a pointed sheaf solid with respect to open embeddings $U \subset V$ of smooth schemes such that the action of G on V is free outside U. Then $\eta_\#(F)$ is solid with respect to open embeddings of smooth schemes.*

Proof. If V' is the open subset of V where the action of G is free, then $U \cup V' = V$ and if $U' := U \cap V'$, a push-out of $U \to V$ is also a push-out of $U' \to V'$: we gained that the action is free everywhere. The next step is applying $\eta_\#$, from pointed sheaves on (QP/G) to pointed sheaves on (QP). This functor is a left adjoint, hence respects colimits and in particular push-outs. It transforms h_U to $h_{U/G}$, and in particular, for $U = S$, the final object into the final object. To check that it respects solidity it is hence sufficient to apply:

Lemma 17. *If G acts freely on U smooth over S, then U/G is smooth.*

Proof. If G is finite etale, for instance S_n, the case which most interests us, this is clear, resulting from $U \to U/G$ being etale. In general one proceeds as follows. The assumption that G acts freely on U implies that U is a G-torsor over U/G. In particular, $U \to U/G$ is faithfully flat. As U is flat over S, this forces U/G to be flat over S. To check smoothness of U/G over S it is hence sufficient to check it geometric fiber by geometric fiber. For $\bar{s}$ a geometric point of S, smoothness of $(U/G)_{\bar{s}}$ amounts to regularity. As $U_{\bar{s}}$ is smooth over $\bar{s}$, hence regular, and $U_{\bar{s}} \to (U/G)_{\bar{s}}$ is faithfully flat, this is [4] (an application of Serre's cohomological criterion for regularity). $\square$

Proposition 45. *The functor $\eta_\#$ respects local (resp. $\mathbf{A}^1$-) equivalences between termwise ind-solid simplicial sheaves.*

Proof. Let $f : F \to F'$ be a local equivalence between termwise ind-solid simplicial sheaves on QP/G. To verify that $\eta_\#(f)$ is a local equivalence it is sufficient to check that for any X in QP and $x \in X$ the map

$$\eta_\#(F)(Spec\mathcal{O}^n_{X,x}) \to \eta_\#(F')(Spec\mathcal{O}^n_{X,x})$$

is a weak equivalence of simplicial sets. Since $\eta_\#$ is a left adjoint, the functor

$$F \mapsto \eta_\#(F)(Spec\mathcal{O}^n_{X,x}) \tag{45.1}$$

commutes with colimits. For an open embedding $U \to V$ in QP/G, $U/G \to V/G$ is again an open embedding and we can apply to (45.1) Theorem 6.

Let $f : F \to F'$ be an $\mathbf{A}^1$-equivalence. Consider the square

$$\begin{array}{ccc} R(F) & \xrightarrow{R(f)} & R(F') \\ \downarrow & & \downarrow \\ F & \xrightarrow{f} & F' \end{array}$$

By the first part of proposition $\eta_\#$ takes the vertical maps to local equivalences. Since $\eta_\#$ commutes with colimits, Lemma 13 implies that $\eta_\#(R(f))$ is in the $\bar{\Delta}$-closure of the class which contains $\eta_\#((K_Q)_+ \to X_+)$ for Q upper distinguished and $\eta_\#((X \times \mathbf{A}^1)_+ \to X_+)$ for X in QP/G. By Theorem 4 it suffice to prove that morphisms of these two types are $\mathbf{A}^1$-equivalences. For morphisms of the first type it follows from the first half of the proposition. For the morphism of the second type it follows from the fact that morphisms $\eta_\#((X \times \mathbf{A}^1)_+ \to X_+)$ and $\eta_\#(X_+ \stackrel{Id\times\{0\}}{\to} (X \times \mathbf{A}^1)_+)$ are mutually inverse $\mathbf{A}^1$-homotopy equivalences. $\square$

Define $\mathbf{L}\eta_\# : Ho_\bullet \to Ho_\bullet$ (and similarly on $Ho_{\mathbf{A}^1,\bullet}$) setting

$$\mathbf{L}\eta_\#(F) := \eta_\#(R(F))$$

where $R(F)$ is defined in Construction 29. Proposition 45 shows that $\mathbf{L}\eta_\#$ is well defined and that for a termwise ind-solid F one has $\mathbf{L}\eta_\#(F) \cong \eta_\#(F)$.

5.2 *The Functor $X \mapsto X^W$*

As in Sect. 3.1, we fix G and S. We also fix W in QP/G which is finite and flat over S.

For F a presheaf on QP/G, we define F^W to be the internal hom object $\underline{Hom}(h_W, F)$. Its value on U is $F(U \times_S W)$. If F is a sheaf, so is F^W.

Example 6. Take G and W deduced from the finite group S_n acting on $\{1, \dots, n\}$ by permutations. In that case, if F is represented by X, with a trivial action of S_n, then F^W is represented by X^n, on which S_n acts by permutation of the factors.

Remark 1. If F is representable (resp. and represented by X smooth over S), so is F^W. More precisely, if F is represented by X in QP/G, consider the contravariant functor on Sch/S of morphisms of schemes from W to X, that is the functor

$$U \mapsto Hom_U(W \times_S U, X \times_S U)$$

This functor is representable, represented by some Y quasi-projective over S (resp. and smooth). This Y carries an obvious action ρ of G, and (Y, ρ) in QP/G represents F^W. Proof: by attaching to a morphism $W \to X$ its graph, one maps the functor considered into the functor of finite subschemes of $W \times_S X$, of the same rank as W, that is the functor

$$U \mapsto \left\{ \begin{array}{c} \text{subschemes of } (W \times_S X) \times_S U \text{ finite and flat over } U, \\ \text{with the same rank as } W \times_S U \text{ over } U. \end{array} \right\}$$

The later functor is represented by a quasi-projective scheme *Hilb*, by the theory of Hilbert schemes. The condition that $\Gamma \subset W \times_S X$ be the graph of a morphism from W to X is an open condition. This means: let $\Gamma \subset (W \times_S X) \times_S U$ be a subscheme finite and flat over U. There is U' open in U such that for any base change $V \to U$, the pull-back Γ_V of Γ is the graph of some V-morphism from $W \times_S V$ to $W \times_S X$ if and only if V maps to U'. This gives the existence of the required Y, and that it is open in *Hilb*. If X is smooth the smoothness of Y follows from the infinitesimal lifting criterion. The quasi-projectivity follows from that of *Hilb*. On the functors represented, the action $g(y) = gyg^{-1}$ of G is clear. For T in QP/G, one has

$$Hom_{QP/G}(T, Y) = Hom_G(T, \underline{Hom}(W, X)) = Hom_G(T \times_S W, X) =$$

$$= Hom_{QP/G}(T \times_S W, X) = F^W(T)$$

Let C be a class of open embeddings in $(QP/G)_{Nis}$. We will simply say "solid" for "C-solid".

Theorem 7. *If F is a solid sheaf on $(QP/G)_{Nis}$, so is F^W.*

If a morphism of sheaves $A \to F$ is open, i.e. relatively representable by open (equivariant) embeddings, there is a natural sequence of sheaves intermediate between A^W and F^W. In the case considered in Example 6, and for $h_U \to h_X$, they are represented by the open equivariant subschemes $(X, U)^n_k$ of X^n consisting of those n-uples $(x_1, \ldots, x_n)$ for which at least k of the x_i are in U. The formal definition is as follows.

A section of F^W over T is a section s of F over $W \times_S T$. Let $U(s)$ be the equivariant open subscheme of $W \times_S T$ on which s is in A. The sheaf $(F, A)^W_k$ is the subsheaf of F^W consisting of those s such that all fibers $U(s)_t$ of $U(s)$ over T are of degree at least k. The condition that the fiber at k be of degree $\geq k$ is open in t, and it follows that the inclusion of $(F, A)^W_k$ in F^W is open. For $k = 0$, $(F, A)^W_k$ is simply F^W. For k large, it is A^W.

Lemma 18. *Suppose that $A \to F$ is deduced by push-out from an open map $B \to G$, so that we have a cocartesian square*

$$\begin{array}{ccc} B & \longrightarrow & G \\ \downarrow & & \downarrow \\ A & \longrightarrow & F \end{array} \tag{45.2}$$

Then, for each k, the cartesian square

$$\begin{array}{ccc} (\text{fiber product}) & \longrightarrow & (A \coprod G, A)_k^W \\ \downarrow & & \downarrow \\ (F, A)_{k+1}^W & \longrightarrow & (F, A)_k^W \end{array} \tag{45.3}$$

is cocartesian as well.

Proof. The site $(QP/G)_{Nis}$ has enough points: as a consequence of Lemma 2, for each X in QP/G and $x \in X/G$, the functor

$$F \mapsto colim \quad F(X \times_{X/G} V),$$

the limit being taken over the Nisnevich neighborhoods of x in X/G, is a point (= a fiber functor). The class of all such functors is clearly conservative. Such a functor depends only on $Y := X \times_{X/G} (X/G)_x^h$, where $(X/G)_x^h$ is the henselization of X/G at x, and Y can be any equivariant S-scheme which is a finite disjoint union of local henselian schemes essentially of finite type over S, and for which Y/G is local. We call such a scheme G-local henselian, and write $F \mapsto F(Y)$ for the corresponding fiber functor.

We will show that (45.3) becomes cocartesian after application of any of the fiber functors $F \mapsto F(Y)$ defined above. It suffices to show that for any s in $(F, A)_k^W(Y)$, the fiber of (45.3)(Y) above s is cocartesian in Set. This fiber is of the form

$$\begin{array}{ccc} K \times L & \longrightarrow & K \\ \downarrow & & \downarrow \\ L & \longrightarrow & \{s\} \end{array}$$

and such a square is cocartesian if and only if whenever K or L is empty, the other is reduced to one element. Here, we also know that $L \to \{s\}$ is a injective. It hence suffice to check that if L is empty, then K is reduced to one element. Fix s in $(F, A)_k^W(Y)$, and view it as a section of F over $W \times_S Y$. Let $U \subset W \times_S Y$ be the open equivariant subset where it is in A. The assumption that s be in $(F, A)_k^W$ means that the degree of the fiber U_g of $U \to Y$ at a closed point y of Y is at least k. By G-equivariance of U, this degree is independent of the chosen y. We have to show that if s is not in $(F, A)_{k+1}^W(Y)$, that is if this degree is exactly k, then s is the image of a unique element of $(A \coprod G, A)_k^W$.

The scheme $W \times_S Y$ is a disjoint union of G-local henselian schemes $(W \times_S Y)_i$. By assumption, (45.2)($(W \times_S Y)_i$) is cocartesian, hence if s is not in A on $(W \times_S Y)_i$, then on $(W \times_S Y)_i$ it comes from a unique $\tilde{s}_i$ in G. Let $(W \times_S Y)'$ be the union of those $(W \times_S Y)_i$ on which s is in A, and $(W \times_S Y)''$ be the union of $(W \times_S Y)_i$ on which it is not. That s is in $(F, A)^W_k$ but not in $(F, A)^W_{k+1}$, means that $(W \times_S Y)'$ is of degree $d = k$ over Y. On $(W \times_S Y)'$, s is in A. On $(W \times_S Y)''$, it comes from a unique $\tilde{s}$ in G. The section

$$s_1 := (s \text{ in } A \text{ on } (W \times_S Y)', \tilde{s} \text{ on } (W \times_S Y)'')$$

of $A \coprod G$ over $W \times_S Y$ is a section of $(A \coprod G, A)^W_k$ on Y lifting s. It is the unique such lifting: any other lifting s_2, viewed as a section of $A \coprod G$ on $W \times_S Y$, can be in A at most on $(W \times_S Y)'$, hence must be in A on the whole of $(W \times_S Y)'$ which has just the required degree over Y. This determines s_2 uniquely on $(W \times_S Y)'$, where it is in A, as well as on $(W \times_S Y)''$, where it is the unique lifting of s to G. $\square$

Proof of Theorem 7: The induction which works to prove Theorem 7 is the following. As F is solid, it sits at the end of a sequence

$$\emptyset \to F_1 \to \ldots \to F_n = F$$

where each $F_i \to F_{i+1}$ is a push-out of some open embedding in QP/G. We prove by induction on i that for any Y, $(F_i \coprod h_Y)^W$ is solid. For $i = 1$, F_1 is representable, hence so is $(F_1 \coprod h_Y)^W$ (1). A fortiori, $(F_1 \coprod h_Y)^W$ is solid. For the induction step one applies the following lemma to $F_i \coprod h_Y \to F_{i+1} \coprod h_Y$.

Lemma 19. *Let*

$$\begin{array}{ccc} h_U & \longrightarrow & h_X \\ \downarrow & & \downarrow \\ F & \longrightarrow & G \end{array}$$

be a cocartesian square with U open in X. Suppose that for any Z, $(F \coprod h_Z)^W$ is solid. Then $F^W \to G^W$ is solid.

Proof. As $F \to G$ is open by Proposition 33, the $(G, F)^W_j$ are defined. It suffices to prove that for each j, the open morphism $(G, F)^W_{j+1} \to (G, F)^W_j$ is solid.

By Lemma 18, this morphism sits in a cartesian and cocartesian square

$$\begin{array}{ccc} (\text{fiber product}) & \xrightarrow{[2]} & (F \coprod h_X, F)^W_k \\ \downarrow & & \downarrow \\ (G, F)^W_{j+1} & \xrightarrow{[1]} & (G, F)^W_j \end{array} \tag{45.4}$$

By assumption, $(F \coprod h_X)^W$ is solid. It follows that $(F \coprod h_X)^W_k$ is solid too (apply Proposition 37 to the open morphism $(F \coprod h_X)^W_k \to (F \coprod h_X)^W$). As [1] is open, so is [2], and by Proposition 36, [2] is solid. The map [1] is then solid as a push-out of a solid map. □

Example 7. It is not always true that if $f : A \to B$ is a solid morphism, so is f^W. Take G the trivial group and W two points (i.e. $S \coprod S$). Then $F^W = F^2$. For any sheaf F, the inclusion f of F in $F \coprod pt$ is solid (deduced by push-out from $\emptyset \to pt$), and applying $(-)^W$, we obtain $F^2 \to F^2 \coprod F \coprod F \coprod pt$. Pulling back by the natural map from F to one of the summands F (an open map), we see that if f^W is solid, so is F.

Corollary 46. *If $f : F \to G$ is open and G solid, then $f^W : F^W \to G^W$ is solid. In particular, if G is pointed solid, so is G^W.*

Proof. f^W is open and one applies Theorem 7 and Proposition 36. □

We now define for pointed sheaves on $(QP/G)_{Nis}$ a "smash" variant of the construction $F \mapsto F^W$. If we assume that the marked point $pt \to F$ is open, it is

$$F^{\wedge W} := F^W/(F, pt)_1^W$$

that is F^W with $(F, pt)_1^W$ contracted to the new base point. This definition can be repeated for any pointed sheaf if, for any monomorphism $A \to F$, we define $(F, A)_1^W$ to be the following subsheaf of F^W: a section s of $F^W(X) = F(X \times W)$ is in $(F, A)_1^W$ if for any non empty $X' \to X$, there exists a commutative diagram

$$\begin{array}{ccc} X'' & \longrightarrow & X \times W \\ \downarrow & & \downarrow \\ X' & \longrightarrow & X \end{array}$$

with X'' non empty and s in A on X''.

Example 8. Under the assumptions of Example 6, if U is open in X with complement Z and if $F = h_X/h_U$, then $F^{\wedge W}$ is $h_{X^n}/h_{X^n - Z^n}$, where X^n has the natural action of S_n. In particular, if F is the Thom space of a vector bundle E over Y (that is, h_E with $h_{E - s_0(Y)}$ contracted to the base point), then $F^{\wedge W}$ is the Thom space of the S_n-equivariant vector bundle $\oplus pr_i^* E$ on Y^n.

Proposition 47. *If a pointed sheaf F is pointed solid, that is if $pt \to F$ is solid, then so is $F^{\wedge W}$.*

Proof. As F^W is solid, the open morphism $(F, pt)_1^W \to F^W$ is solid too (Proposition 36), and $pt \to F^{\wedge W}$ is deduced from it by push-out. □

The definition of $F^{\wedge W}$ immediately implies the following:

Lemma 20. *Let Y be a G-local henselian scheme. Then*

$$F^{\wedge W}(Y) = \bigwedge_i F((W \times Y)_i)$$

where $(W \times Y)_i$ are G-local henselian schemes such that

$$W \times Y = \coprod_i (W \times Y)_i$$

Proposition 48. *The functor $F^{\wedge W}$ respects local and $\mathbf{A}^1$-equivalences.*

Proof. Let $f : F \to H$ be a local equivalence. To check that $F^W \to H^W$ is a local equivalence it is enough to show that for any G-local henselian Y, the map

$$F^W(Y) = F(Y \times W) \to H(Y \times W) = H^W(Y)$$

is a weak equivalence of simplicial sets. This follows from Lemma 20.

Let f be an $\mathbf{A}^1$-equivalence. By the first part it is sufficient to show that $R(f)^W : R(F)^W \to R(H)^W$ is an $\mathbf{A}^1$-equivalence. We use the characterization of $\mathbf{A}^1$-equivalences given in Theorem 5. Since $F \mapsto F^{\wedge W}$ commutes with filtering colimits and preserves local equivalences it suffices to check that it takes $\mathbf{A}^1$-homotopy equivalences to $\mathbf{A}^1$-homotopy equivalences. This is seen using the natural map

$$F^{\wedge W} \wedge (h_{\mathbf{A}^1})_+ \to (F \wedge (h_{\mathbf{A}^1})_+)^{\wedge W}.$$

□

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